GEBZE TECHNICAL UNIVERSITY

Computer Engineering Department

CSE 396 Module Documentation

Çılgın Kurye

Module M3: Physics and Collision

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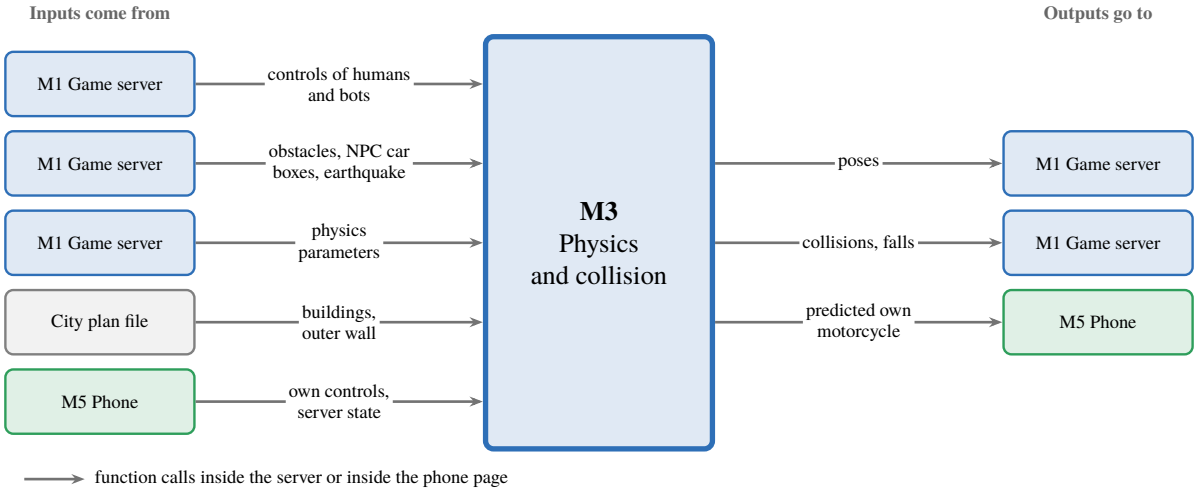
1. Purpose

M3 moves the motorcycles. It turns throttle, brake and steering into speed, direction and position. It finds collisions, decides when a motorcycle falls, and adds the effect of obstacles and earthquakes to the motion. The same code runs in two places: on the server, where its result is the real one, and on the phone, where it guesses the player's own motorcycle before the server answers.

Runs on	Technology	Requirements
Main computer inside the game server, and the phone browser for prediction only	TypeScript library with a 3D physics engine that runs on the web, on the server and in the phone browser	R3 R4 R6 R7 R11

M3 has no network code. On the server it talks only to M1, and on the phone only to M5. The figure shows both uses.



2. Requirements covered

M3 takes part in five requirements. The last column says what will work when M3 is finished.

ID	Requirement	What M3 does for it	When M3 is done
R3	Driving and controls	Throttle, brake, left and right. No reverse. A U turn by steering in a small circle at low speed	The motorcycle drives, brakes and turns like an arcade game
R4	Collisions and falls	Collisions with buildings, barriers, NPC cars and motorcycles. A fast hit makes the motorcycle fall	Nothing drives through a wall, and fast crashes end in a fall
R6	Obstacles	Road work and accidents are solid. An accident has a slow zone. A puddle makes the motorcycle slide	Each obstacle type feels different when you drive
R7	Earthquake	Less grip inside the earthquake area that M1 sends	Riding in an earthquake is harder
R11	Engineering requirements	The same input always gives the same result. The server result wins, the phone only predicts	The own motorcycle reacts at once and still matches the server

3. Inputs and outputs

M1 calls M3 two times in every tick with the inputs below. M5 uses the same code for one motorcycle.

Direction	Data	From / to	Content
In	City plan	Shared file	Roads, lanes, blocks and buildings as fixed colliders
In	Controls	M1	Every tick: throttle and brake from 0 to 1, steer from −1 to 1, for humans and bots, and a speed factor for each player
In	Obstacles, NPC cars	M1	Obstacle type, road, lane, position, and the boxes of the NPC cars that M1 gets from M4
In	Earthquake	M1	The earthquake area and the motorcycles that fall
In	Parameters	M6, through M1	Top speed, grip, fall speed, grip in a puddle and in an earthquake
Out	Poses	M1	Position, direction, speed, lean, sliding, fallen
Out	Events	M1	Collisions (with what and how fast) and falls. M1 gives the car hits to M4
Out	Predicted pose	M5	The player's own motorcycle before the server answers

4. Main components

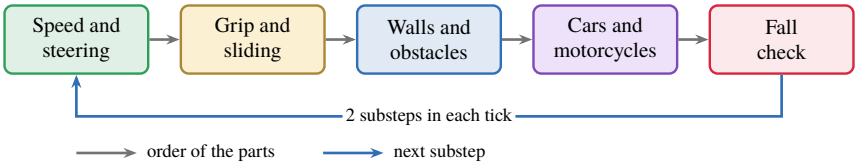
M3 has five parts. The first four run on the server, the last one runs on the phone.

Component	Takes	Gives
Motorcycle model	Controls, parameters	New speed, direction, lean and sliding
Colliders	City plan, obstacle list, NPC cars	Solid shapes, slow zones and slippery areas
Collision solver and fall judge	Moved objects, colliders	Pushed apart objects, collision and fall events
Earthquake effect	Earthquake area	Lower grip inside the area
Predictor	Own controls, server state	Predicted own motorcycle with a smooth correction

5. How it works

One physics step:

The server runs 30 ticks per second, and M3 runs 2 substeps in each tick. Small steps keep fast motorcycles from passing through thin walls. Each substep does five things in a fixed order, as the figure shows. Because the order is fixed, the same inputs always give the same result on the server and on the phone.

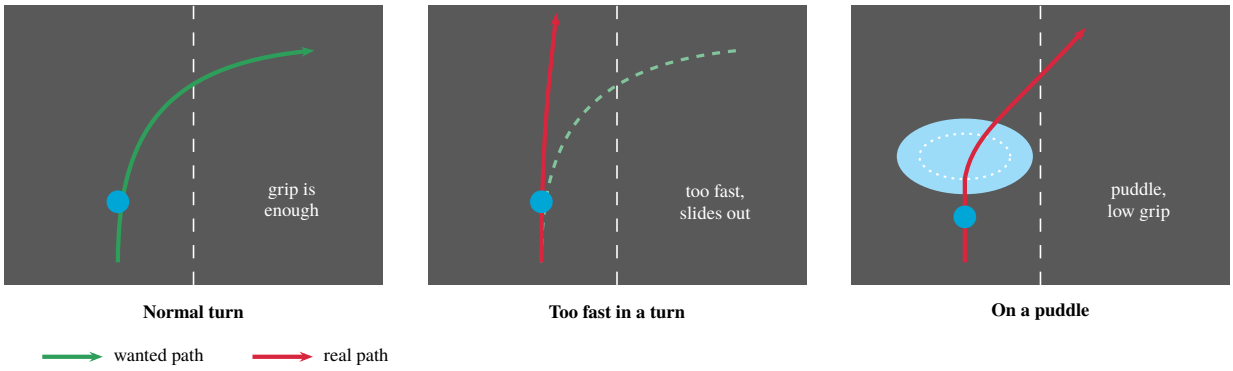


Driving:

The throttle adds speed up to the top speed, and the brake and air drag take speed away. There is no reverse. The steering changes the direction. At low speed the motorcycle can turn in a small circle, so a U turn is easy. At high speed the turn is wider. The model also gives a lean angle, which M5 uses to tilt the 3D motorcycle in a turn.

Grip and sliding:

In a turn the motorcycle needs a side force. M3 compares the size of this force with the grip limit. If the force is below the limit, the motorcycle follows the turn. If it is above the limit, the motorcycle slides out of the turn. We compare only the size, so a left turn and a right turn act the same. On a water puddle the grip limit is very low, so the motorcycle slides but does not slow down. Inside the earthquake area the grip is lower too.

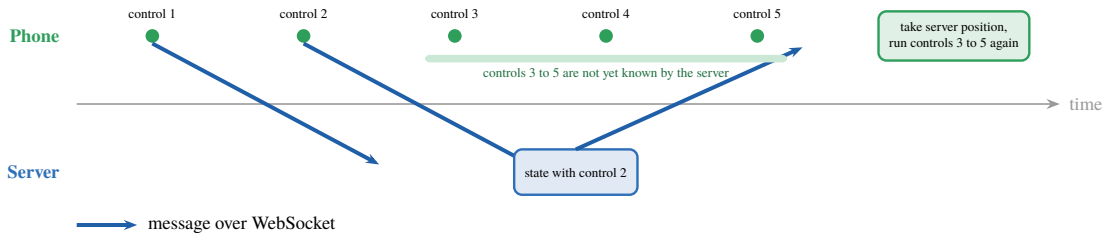


Collisions and falls:

Buildings, the outer wall and the road work barriers come from the city plan and the obstacle list as solid shapes. NPC cars and motorcycles have simple shapes too. A contact pushes the objects apart. If the hit speed is above the fall speed that the admin set, the motorcycle falls. When two motorcycles hit each other and their relative speed is above the limit, both of them fall. A fallen motorcycle is not solid, so it does not block the road for others. M1 gives the penalty time and later asks M3 to stand the motorcycle up again. An accident is solid in one lane and has a slow zone around it.

Prediction on the phone:

On the phone, the player must see the own motorcycle move at once, not after the delay of the network. So M5 runs a copy of M3 for this one motorcycle. Each control gets a number and is kept in a list. When a server state arrives, the phone takes the real position from the server, throws away the controls that the server has already used, and runs the remaining controls again. If the new result is close to what the player sees, the difference is blended away in a short time. If it is far, for example after a crash, the motorcycle jumps to the right place.



6. Testing

M3 needs no other module. A test makes a world from the city plan and gives scripted controls, fixed obstacles and fixed NPC car boxes as test data. We check a straight run and braking, a U turn, slides to the left and to the right, wall and car crashes, the earthquake radius, and that the same inputs give exactly the same result two times. The predictor is tested with recorded server states and added delay and loss. A small test page shows a simple 3D view of the city, so one motorcycle can also be driven with the keyboard and the parameters can be changed with keys.